

Week 4

4.1 Gradient and Directional Derivatives

Definition 4.18. Let $\Omega \subseteq \mathbb{R}^n$ be open, $\vec{a} \in \Omega$, $f : \Omega \rightarrow \mathbb{R}$. We define the gradient vector of f at $\vec{a}$ to be

$$\nabla f(\vec{a}) = \left(\frac{\partial f}{\partial x_1}(\vec{a}), \dots, \frac{\partial f}{\partial x_n}(\vec{a}) \right) \in \mathbb{R}^n.$$

Example 90. If $f(x, y) = x^2 + 2xy$, then its gradient is given by

$$\nabla f(x, y) = (f_x, f_y) = (2x + 2y, 2x),$$

and so at the point $(1, 2) \in \mathbb{R}^2$, its gradient is the vector $\nabla f(1, 2) = (6, 2) \in \mathbb{R}^2$.

If a function f is differentiable at a point $\vec{a}$, then its affine approximation at that point can be expressed more succinctly:

$$\begin{aligned} L(\vec{x}) &= f(\vec{a}) + \sum_{i=1}^n \frac{\partial f}{\partial x_i}(\vec{a})(x_i - a_i) \\ &= f(\vec{a}) + \nabla f(\vec{a}) \cdot (\vec{x} - \vec{a}). \end{aligned}$$

Definition 4.19. Let $\Omega \subseteq \mathbb{R}^n$ be open, $\vec{a} \in \Omega$, $f : \Omega \rightarrow \mathbb{R}$. For any unit vector $\vec{u} \in \mathbb{R}^n$ ($\|\vec{u}\| = 1$), the directional derivative of f in the direction $\vec{u}$ at $\vec{a}$ is

$$D_{\vec{u}}f(\vec{a}) = \lim_{h \rightarrow 0} \frac{f(\vec{a} + h\vec{u}) - f(\vec{a})}{h},$$

i.e., the rate of change of f in direction $\vec{u}$ at the point $\vec{a}$.

Example 91. Choosing $\vec{u} = \vec{e}_i \in \mathbb{R}^n$, we recover the partial derivative $D_{\vec{e}_i}f(\vec{a}) = \frac{\partial f}{\partial x_i}(\vec{a})$, for any $i = 1, \dots, n$.

Lemma 92. If f is differentiable at $\vec{a}$, and $\vec{u} \in \mathbb{R}^n$ is a unit vector, then

$$D_{\vec{u}}f(\vec{a}) = \nabla f(\vec{a}) \cdot \vec{u}.$$

Proof. Since f is differentiable at $\vec{a}$, we have

$$f(\vec{x}) = f(\vec{a}) + \nabla f(\vec{a}) \cdot (\vec{x} - \vec{a}) + \varepsilon(x), \tag{4.11}$$

for some error function $\varepsilon(x)$ satisfying $\lim_{\vec{x} \rightarrow \vec{a}} \frac{\varepsilon(\vec{x})}{\|\vec{x} - \vec{a}\|} = 0$. Setting $\vec{x} = \vec{a} + h\vec{u}$ in (4.11), we find that

$$f(\vec{a} + h\vec{u}) - f(\vec{a}) = h(\nabla f(\vec{a}) \cdot \vec{u}) + \varepsilon(\vec{a} + h\vec{u}),$$

and therefore

$$\begin{aligned} D_{\vec{u}}f(\vec{a}) &= \lim_{h \rightarrow 0} \frac{f(\vec{a} + h\vec{u}) - f(\vec{a})}{h} \\ &= \nabla f(\vec{a}) \cdot \vec{u} + \lim_{h \rightarrow 0} \frac{\varepsilon(\vec{a} + h\vec{u})}{h} = \nabla f(\vec{a}) \cdot \vec{u}. \end{aligned} \quad \square$$

For a non-zero vector $\vec{v} \in \mathbb{R}^n$, then the direction of $\vec{v}$ is the unit vector $\frac{\vec{v}}{\|\vec{v}\|}$.

Example 93. Let $f(x, y) = \arcsin(\frac{x}{y})$. Find the rate of change of f at the point $(1, \sqrt{2})$ in the direction of $\vec{v} = (1, -1)$.

Solution: Set $\vec{u} = \frac{\vec{v}}{\|\vec{v}\|} = (\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}})$ to be the direction of $\vec{v}$. We note that

$$f_x(x, y) = \frac{1}{\sqrt{y^2 - x^2}}, \quad f_y(x, y) = \frac{-x}{y\sqrt{y^2 - x^2}},$$

and so f, f_x, f_y are all continuous near the point $(1, \sqrt{2})$. In particular f is C^1 near $(1, \sqrt{2})$ and hence f is differentiable at $(1, \sqrt{2})$.

We can therefore use the gradient of f to calculate the desired directional derivative

$$\begin{aligned} D_{\vec{u}}f((1, \sqrt{2})) &= \nabla f((1, \sqrt{2})) \cdot \vec{u} \\ &= (f_x(1, \sqrt{2}), f_y(1, \sqrt{2})) \cdot (\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}}) \\ &= \left(1, -\frac{1}{\sqrt{2}}\right) \cdot \left(\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}}\right) = \frac{1}{\sqrt{2}} + \frac{1}{2}. \end{aligned}$$

4.1.1 Geometric Meaning of the Gradient

If f is differentiable at $\vec{a}$, $\vec{u} \in \mathbb{R}^n$ a unit vector, then

$$D_{\vec{u}}f(\vec{a}) = \nabla f(\vec{a}) \cdot \vec{u}.$$

By the Cauchy-Schwarz inequality,

$$|\nabla f(\vec{a}) \cdot \vec{u}| \leq \|\nabla f(\vec{a})\| \cdot \underbrace{\|\vec{u}\|}_{=1} = \|\nabla f(\vec{a})\|.$$

Moreover, if $\nabla f(\vec{a}) \neq 0$, then

$$-\|\nabla f(\vec{a})\| \leq \nabla f(\vec{a}) \cdot \vec{u} \leq \|\nabla f(\vec{a})\|,$$

where equality holds in the first inequality iff $\nabla f(\vec{a}) = -\lambda\vec{u}$ for some $\lambda > 0$, and equality holds in the second inequality iff $\nabla f(\vec{a}) = \lambda\vec{u}$ for some $\lambda > 0$.

Indeed, at $\vec{a}$, $f(\vec{x})$ increases (decreases) most rapidly in the direction of $\nabla f(\vec{a})$ ($-\nabla f(\vec{a})$) at a rate of $\|\nabla f(\vec{a})\|$.

Contents

Remark. When we defined directional derivatives, we specified that $\vec{u}$ was a unit vector. We could have instead defined a directional derivative for a general vector $\vec{v} \in \mathbb{R}^n$ in the exact same way

$$D_{\vec{v}}f(\vec{a}) := \lim_{h \rightarrow 0} \frac{f(\vec{a} + h\vec{v}) - f(\vec{a})}{h}.$$

We note that

$$D_{\vec{v}}f(\vec{a}) = \begin{cases} \|\vec{v}\| D_{\frac{\vec{v}}{\|\vec{v}\|}}f(\vec{a}) & : v \neq 0 \\ 0 & : \vec{v} = 0 \end{cases},$$

and therefore, we lose no information by restricting our definition of directional derivatives to the case where v is a unit vector.

4.1.2 Summary

Suppose $f : \mathbb{R}^n \rightarrow \mathbb{R}$, $\vec{a} \in \mathbb{R}^n$.

Types of Derivatives:

- Directional: $D_{\vec{u}}f(\vec{a}) = \lim_{h \rightarrow 0} \frac{f(\vec{a} + h\vec{u}) - f(\vec{a})}{h}$, with $\|\vec{u}\| = 1$.
- Partial: $\frac{\partial f}{\partial x_i}(\vec{a}) = D_{\vec{e}_i}f(\vec{a})$, for the standard basis $\vec{e}_i \in \mathbb{R}^n$.
- Gradient: $\nabla f(\vec{a}) = \left(\frac{\partial f}{\partial x_1}(\vec{a}), \dots, \frac{\partial f}{\partial x_n}(\vec{a}) \right) \in \mathbb{R}^n$.
- Higher order: e.g. $\frac{\partial^2 f}{\partial x_j \partial x_i}(\vec{a}) = f_{x_i x_j}(\vec{a})$.

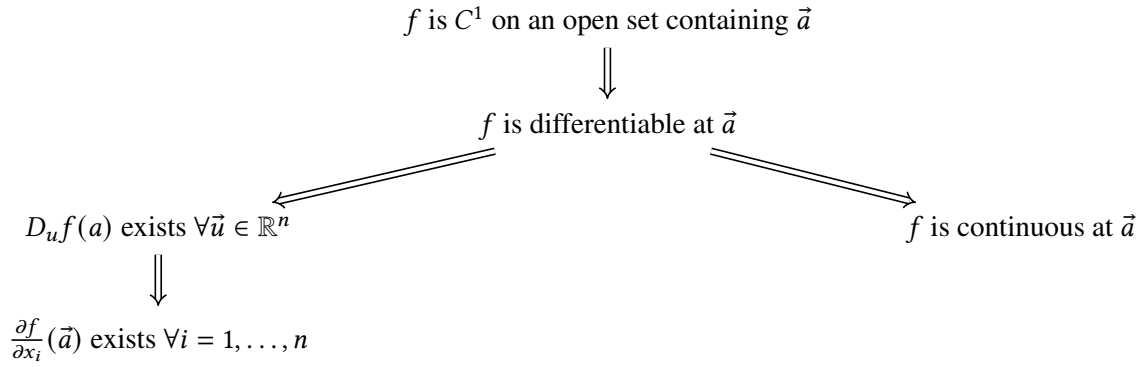
f is a C^k -function if f and all of its partial derivatives up to (and including) order k exist and are continuous.

Affine Approximation:

- $L(\vec{x}) = f(\vec{a}) + \nabla f(\vec{a}) \cdot (\vec{x} - \vec{a})$.
- $\varepsilon(\vec{x}) = f(\vec{x}) - L(\vec{x})$
- $f(\vec{x})$ is differentiable at $\vec{a}$ if $\lim_{\vec{x} \rightarrow \vec{a}} \varepsilon(\vec{x}) (\|\vec{x} - \vec{a}\|)^{-1} = 0$.

Relations Between Derivatives:

$$C^\infty \supsetneq \dots \supsetneq C^{k+1} \supsetneq C^k \supsetneq \dots \supsetneq C^1 \supsetneq C^0.$$



Exercise: Find functions $f : \mathbb{R}^n \rightarrow \mathbb{R}$ such that

- f is continuous on $\mathbb{R}^n$, but at some point $\vec{a} \in \mathbb{R}^n$, none of the partial derivatives $\frac{\partial f}{\partial x_i}(\vec{a})$ exist.
- At some point $\vec{a} \in \mathbb{R}^n$, $D_{\vec{u}}f(\vec{a})$ exists for all unit vectors $\vec{u}$, but f is not continuous at $\vec{a}$.
- At some point $\vec{a} \in \mathbb{R}^n$, all of the partial derivatives $\frac{\partial f}{\partial x_i}(\vec{a})$ exist, but not every directional derivative $D_{\vec{u}}f(\vec{a})$ exists.

4.2 Differentiability for Multivariable Functions

We will now generalise our discussion on affine approximations and differentiability to multivariable, *vector-valued* functions, i.e. functions $\vec{f} : \Omega \subset \mathbb{R}^n \rightarrow \mathbb{R}^m$.

4.2.1 Matrix Multiplication

We begin with a brief recap on matrix multiplication. Let A be an $k \times m$ matrix, and B a $m \times n$ matrix. Then we can multiply the two matrices to get the $k \times n$ matrix AB . More explicitly, let

$$A = \begin{pmatrix} a_{11} & \cdots & a_{1m} \\ \vdots & \ddots & \vdots \\ a_{k1} & \cdots & a_{km} \end{pmatrix}$$

which we view as a linear map from $\mathbb{R}^m$ to $\mathbb{R}^k$. That is, given a vector $x \in \mathbb{R}^m$, we have

$$Ax = \begin{pmatrix} a_{11} & \cdots & a_{1m} \\ \vdots & \ddots & \vdots \\ a_{k1} & \cdots & a_{km} \end{pmatrix} \begin{pmatrix} x_1 \\ \vdots \\ x_m \end{pmatrix} = \begin{pmatrix} a_{11}x_1 + \cdots + a_{1m}x_m \\ \vdots \\ a_{k1}x_1 + \cdots + a_{km}x_m \end{pmatrix} \in \mathbb{R}^k. \quad (4.12)$$

Choosing the row vectors

$$\alpha_i = (a_{i1}, \dots, a_{im}) \in \mathbb{R}^m, \quad \forall 1 \leq i \leq k,$$

we can rewrite (4.12) as

$$Ax = \begin{pmatrix} -\alpha_1 - \\ \vdots \\ -\alpha_k - \end{pmatrix} \begin{pmatrix} | \\ x \\ | \end{pmatrix} = \begin{pmatrix} \alpha_1 \cdot x \\ \vdots \\ \alpha_k \cdot x \end{pmatrix} \in \mathbb{R}^k.$$

Similarly, given a row vector $y = (y_1, \dots, y_m) \in \mathbb{R}^m$, we can multiply B on the left by y to get a vector in $\mathbb{R}^n$.

$$yB = (y_1 \quad \cdots \quad y_m) \begin{pmatrix} b_{11} & \cdots & b_{1n} \\ \vdots & \ddots & \vdots \\ b_{m1} & \cdots & b_{mn} \end{pmatrix} = (y_1 b_{11} + \cdots + y_m b_{m1}, \dots, y_1 b_{1n} + \cdots + y_m b_{mn}) \in \mathbb{R}^n. \quad (4.13)$$

Choosing now the column vectors

$$\beta_j = \begin{pmatrix} b_{1j} \\ \vdots \\ b_{mj} \end{pmatrix} \in \mathbb{R}^m, \quad \forall 1 \leq j \leq n,$$

we can rewrite (4.13) as

$$yB = (-y-) \begin{pmatrix} | & & | \\ \beta_1 & \cdots & \beta_n \\ | & & | \end{pmatrix} = (y \cdot \beta_1, \dots, y \cdot \beta_n) \in \mathbb{R}^n.$$

Therefore, we can write the matrix product AB as

$$AB = \begin{pmatrix} a_{11} & \cdots & a_{1m} \\ \vdots & \ddots & \vdots \\ a_{k1} & \cdots & a_{km} \end{pmatrix} \begin{pmatrix} b_{11} & \cdots & b_{1n} \\ \vdots & \ddots & \vdots \\ b_{m1} & \cdots & b_{mn} \end{pmatrix} = \begin{pmatrix} -\alpha_1 - \\ \vdots \\ -\alpha_k - \end{pmatrix} \begin{pmatrix} | & & | \\ \beta_1 & \cdots & \beta_n \\ | & & | \end{pmatrix} = \begin{pmatrix} \alpha_1 \cdot \beta_1 & \cdots & \alpha_1 \cdot \beta_n \\ \vdots & \ddots & \vdots \\ \alpha_k \cdot \beta_1 & \cdots & \alpha_k \cdot \beta_n \end{pmatrix}$$

Example 94.

$$\begin{aligned} \overbrace{\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}}^A \overbrace{\begin{pmatrix} 5 & 6 & 7 \\ 8 & 9 & 10 \end{pmatrix}}^B &= \begin{pmatrix} (1, 2) \cdot (5, 8) & (1, 2) \cdot (6, 9) & (1, 2) \cdot (7, 10) \\ (3, 4) \cdot (5, 8) & (3, 4) \cdot (6, 9) & (3, 4) \cdot (7, 10) \end{pmatrix} \\ &= \underbrace{\begin{pmatrix} 21 & 24 & 27 \\ 47 & 54 & 61 \end{pmatrix}}_{AB} \end{aligned}$$

4.2.2 The Jacobean Matrix

Given a multivariable function $\vec{f} : \Omega \subseteq \mathbb{R}^n \rightarrow \mathbb{R}^m$, we know that we can reduce its complexity by studying its component functions

$$\vec{f}(\vec{x}) = \begin{pmatrix} f_1(\vec{x}) \\ \vdots \\ f_m(\vec{x}) \end{pmatrix} \in \mathbb{R}^m,$$

Contents

with $f_i : \Omega \rightarrow \mathbb{R}$, for $1 \leq i \leq m$. We now suppose that $\frac{\partial f_i}{\partial x_j}(\vec{a})$ exist for each $i \in \{1, \dots, m\}$ and $j \in \{1, \dots, n\}$, for some $\vec{a} \in \Omega$.

For any fixed $1 \leq i \leq m$, we note that

$$f_i(\vec{x}) = f_i(\vec{a}) + \underbrace{\nabla f_i(\vec{a}) \cdot (\vec{x} - \vec{a})}_{\substack{1 \times n \quad n \times 1}} + \varepsilon_i(\vec{x}),$$

where we regard the gradient as a row vector and $(\vec{x} - \vec{a})$ as a column vector in order to use matrix multiplication. In particular, we can gather these m equations together in the form

$$\begin{pmatrix} f_1(\vec{x}) \\ \vdots \\ f_m(\vec{x}) \end{pmatrix} = \begin{pmatrix} f_1(\vec{a}) \\ \vdots \\ f_m(\vec{a}) \end{pmatrix} + \underbrace{\begin{pmatrix} -\nabla f_1(\vec{a}) - \\ \vdots \\ -\nabla f_m(\vec{a}) - \end{pmatrix}}_{m \times n} \underbrace{\begin{pmatrix} x_1 - a_1 \\ \vdots \\ x_n - a_n \end{pmatrix}}_{n \times 1} + \begin{pmatrix} \varepsilon_1(\vec{x}) \\ \vdots \\ \varepsilon_m(\vec{x}) \end{pmatrix} \in \mathbb{R}^m.$$

Definition 4.20. We define the Jacobian matrix of $\vec{f}$ at $\vec{a}$ to be the $m \times n$ matrix

$$D\vec{f}(\vec{a}) = \begin{pmatrix} \frac{\partial f_1}{\partial x_1}(\vec{a}) & \cdots & \frac{\partial f_1}{\partial x_n}(\vec{a}) \\ \vdots & \ddots & \vdots \\ \frac{\partial f_m}{\partial x_1}(\vec{a}) & \cdots & \frac{\partial f_m}{\partial x_n}(\vec{a}) \end{pmatrix} = \begin{pmatrix} -\nabla f_1(\vec{a}) - \\ \vdots \\ -\nabla f_m(\vec{a}) - \end{pmatrix},$$

and affine approximation of $\vec{f}$ at $\vec{a}$ to be $\vec{L} : \mathbb{R}^n \rightarrow \mathbb{R}^m$, given by

$$\vec{L}(\vec{x}) = \vec{f}(\vec{a}) + D\vec{f}(\vec{a})(\vec{x} - \vec{a}) \in \mathbb{R}^m, \quad \forall \vec{x} \in \mathbb{R}^n.$$

Moreover, we say that $\vec{f}$ is differentiable at $\vec{a}$ if the error term $\vec{\varepsilon}(\vec{x}) := \vec{f}(\vec{x}) - \vec{L}(\vec{x})$ satisfies

$$\lim_{\vec{x} \rightarrow \vec{a}} \frac{\|\vec{\varepsilon}(\vec{x})\|}{\|\vec{x} - \vec{a}\|} = 0.$$

Remark. Note that if $\vec{f}$ is real valued ($m = 1$), then $Df(\vec{a}) = \nabla f(\vec{a})$. Also, from our discussion of limits of vector-valued functions

$$\lim_{\vec{x} \rightarrow \vec{a}} \frac{\|\vec{\varepsilon}(\vec{x})\|}{\|\vec{x} - \vec{a}\|} = 0 \iff \lim_{\vec{x} \rightarrow \vec{a}} \frac{\varepsilon_i(\vec{x})}{\|\vec{x} - \vec{a}\|} = 0, \quad \forall i \in \{1, \dots, m\}.$$

So $\vec{f}$ is differentiable at $\vec{a}$ if and only if all of its components f_i are differentiable at $\vec{a}$.

Employing the language of ‘change vectors’ $\delta \vec{f}$, $\delta \vec{x}$, we have

$$\underbrace{\vec{f}(\vec{x}) - \vec{f}(\vec{a})}_{\Delta \vec{f}} \approx D\vec{f}(\vec{a}) \cdot \underbrace{(\vec{x} - \vec{a})}_{\Delta \vec{x}}$$

This is the multivariable generalisation of the fact that the derivative approximates the difference quotient of a function $f : \mathbb{R} \rightarrow \mathbb{R}$:

$$\frac{f(x) - f(a)}{x - a} \approx f'(a).$$

By the above equation, we see that the Jacobian can be thought of as a linear map $D\vec{f}(\vec{a}) : \mathbb{R}^n \rightarrow \mathbb{R}^m$, mapping $\Delta\vec{x} \in \mathbb{R}^n$ to $\Delta\vec{f} \in \mathbb{R}^m$.

Example 95. Let $\vec{f}(x, y) = ((y + 1) \log x, x^2 - \sin y + 1)$.

- a) Find $D\vec{f}(1, 0)$.
- b) Approximate $\vec{f}(0.9, 0.1)$.

Solution:

- a) Calculating partial derivatives we have

$$D\vec{f} = \begin{pmatrix} \frac{\partial f_1}{\partial x} & \frac{\partial f_1}{\partial y} \\ \frac{\partial f_2}{\partial x} & \frac{\partial f_2}{\partial y} \end{pmatrix} = \begin{pmatrix} \frac{y+1}{x} & \log(x) \\ 2x & -\cos y \end{pmatrix}.$$

Therefore

$$D\vec{f}(1, 0) = \begin{pmatrix} 1 & 0 \\ 2 & -1 \end{pmatrix}.$$

- b)

$$\begin{aligned} \vec{L}(x, y) &= \vec{f}(1, 0) + D\vec{f}(1, 0) \begin{pmatrix} x - 1 \\ y - 0 \end{pmatrix} \\ &= \begin{pmatrix} 0 \\ 2 \end{pmatrix} + \begin{pmatrix} 1 & 0 \\ 2 & -1 \end{pmatrix} \begin{pmatrix} x - 1 \\ y \end{pmatrix} \\ &= \begin{pmatrix} x - 1 \\ 2x - y \end{pmatrix}. \end{aligned}$$

Therefore

$$\vec{f}(0.9, 0.1) \approx \vec{L}(0.9, 0.1) = \begin{pmatrix} -0.1 \\ 1.7 \end{pmatrix}.$$

4.3 Chain Rule

Recall, for two differentiable functions $f, g : \mathbb{R} \rightarrow \mathbb{R}$ of one variable, the chain rule states that

$$(g \circ f)'(x) = g'(f(x)) \cdot f'(x),$$

or, alternatively, if we label $y = f(x)$ and $z = g(y)$, then

$$\frac{dz}{dx} = \frac{dz}{dy} \cdot \frac{dy}{dx}.$$

In other words, *the derivative of a composition of functions is the product of the derivatives.*

Example 96. If $f(x) = x^2$ and $g(x) = 2x + 1$, then $g \circ f(x) = 2x^2 + 1$. In particular,

$$(g \circ f)'(x) = 4x = 2 \cdot 2x = g'(f(x)) \cdot f'(x).$$

For multivariable functions, a similar expression holds between the Jacobian matrices, where instead of a product we have matrix multiplication.

Theorem 97 (Chain Rule). *Let $\vec{f} : \Omega_1 \subseteq \mathbb{R}^n \rightarrow \mathbb{R}^m$, $\vec{g} : \Omega_2 \subseteq \mathbb{R}^m \rightarrow \mathbb{R}^k$, with Ω_1, Ω_2 open. Suppose that $\vec{f}$ is differentiable at $\vec{a} \in \Omega_1$ and $\vec{g}$ is differentiable at $\vec{b} = \vec{f}(\vec{a}) \in \Omega_2$. Then $\vec{g} \circ \vec{f}$ is differentiable at $\vec{a}$ with*

$$\underbrace{D(\vec{g} \circ \vec{f})(\vec{a})}_{k \times n} = \underbrace{D\vec{g}(\vec{f}(\vec{a}))}_{k \times m} \cdot \underbrace{D\vec{f}(\vec{a})}_{m \times n}.$$

In other words, the derivative (Jacobian matrix) of a composition of multivariable functions is the matrix product of the derivatives (Jacobian matrices).

Example 98. Let $\vec{f} : \mathbb{R} \rightarrow \mathbb{R}^2$, $\vec{g} : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be defined by

$$\vec{f}(\theta) = (\cos \theta, \sin \theta), \quad \vec{g}(u, v) = (2uv, u^2 - v^2).$$

Note that $\vec{g} \circ \vec{f} : \mathbb{R} \rightarrow \mathbb{R}^2$, so that $D(\vec{g} \circ \vec{f})(\theta)$ is a 2×1 matrix for any value of $\theta \in \mathbb{R}$. There are two potential ways to calculate the derivative of this composition

Method 1: Find the composition explicitly.

$$\begin{aligned} \vec{g} \circ \vec{f}(\theta) &= \vec{g}(\cos \theta, \sin \theta) \\ &= (2 \cos \theta \sin \theta, \cos^2 \theta - \sin^2 \theta) \\ &= (\sin 2\theta, \cos 2\theta). \end{aligned}$$

Therefore

$$D(\vec{g} \circ \vec{f})(\theta) = \begin{pmatrix} 2 \cos 2\theta \\ -2 \sin 2\theta \end{pmatrix}.$$

Method 2: Use the Chain Rule.

$$\begin{aligned} D\vec{f}(\theta) &= \begin{pmatrix} -\sin \theta \\ \cos \theta \end{pmatrix}, \\ D\vec{g}(u, v) &= \begin{pmatrix} 2v & 2u \\ 2u & -2v \end{pmatrix}, \\ D\vec{g}(\vec{f}(\theta)) &= \begin{pmatrix} 2 \sin \theta & 2 \cos \theta \\ 2 \cos \theta & -2 \sin \theta \end{pmatrix}. \end{aligned}$$

By the Chain Rule

$$\begin{aligned}
 D(\vec{g} \circ \vec{f})(\theta) &= D\vec{g}(\vec{f}(\theta)) \circ D\vec{f}(\theta) \\
 &= \begin{pmatrix} 2 \sin \theta & 2 \cos \theta \\ 2 \cos \theta & -2 \sin \theta \end{pmatrix} \cdot \begin{pmatrix} -\sin \theta \\ \cos \theta \end{pmatrix} \\
 &= \begin{pmatrix} -2 \sin^2 \theta + 2 \cos^2 \theta \\ -4 \cos \theta \sin \theta \end{pmatrix} \\
 &= \begin{pmatrix} 2 \cos 2\theta \\ -2 \sin 2\theta \end{pmatrix}.
 \end{aligned}$$

Example 99. If $\vec{f}(x, y) = (x^2, 3xy, x + y^2)$ and $g(u, v, w) = \frac{uw}{v}$, then

$$\begin{aligned}
 D\vec{f}(x, y) &= \begin{pmatrix} 2x & 0 \\ 3y & 3x \\ 1 & 2y \end{pmatrix}, \\
 Dg(u, v, w) &= \left(\frac{w}{v}, \frac{-uw}{v^2}, \frac{u}{v} \right),
 \end{aligned}$$

Therefore,

$$D\vec{f}(1, 1) = \begin{pmatrix} 2 & 0 \\ 3 & 3 \\ 1 & 2 \end{pmatrix}, \quad Dg(\vec{f}(1, 1)) = Dg(1, 3, 2) = \left(\frac{2}{3}, \frac{-2}{9}, \frac{1}{3} \right),$$

and

$$D(g \circ \vec{f})(1, 1) = Dg(\vec{f}(1, 1)) \cdot D\vec{f}(1, 1) = \left(\frac{2}{3}, \frac{-2}{9}, \frac{1}{3} \right) \begin{pmatrix} 2 & 0 \\ 3 & 3 \\ 1 & 2 \end{pmatrix} = (1, 0). \quad (4.14)$$

In the previous example, we could regard $\vec{f}$ as a change of variables from x, y to u, v, w , and hence we could view the function $g \circ \vec{f}$ as expressing g as a function of x and y . In particular, by an *abuse of notation*, we could read off (4.14) as

$$\frac{\partial g}{\partial x}(1, 1) = 1, \quad \frac{\partial g}{\partial y}(1, 1) = 0.$$

That is, in classical notation, we can express the chain rule as follows:

$$\begin{aligned}
 \frac{\partial g}{\partial x} &= \frac{\partial g}{\partial u} \cdot \frac{\partial u}{\partial x} + \frac{\partial g}{\partial v} \cdot \frac{\partial v}{\partial x} + \frac{\partial g}{\partial w} \cdot \frac{\partial w}{\partial x} \\
 \frac{\partial g}{\partial y} &= \frac{\partial g}{\partial u} \cdot \frac{\partial u}{\partial y} + \frac{\partial g}{\partial v} \cdot \frac{\partial v}{\partial y} + \frac{\partial g}{\partial w} \cdot \frac{\partial w}{\partial y}
 \end{aligned}$$

Example 100. Let $w(x, y, z) = \sqrt{x^2 + y^2 + z^2}$, and

$$x = 3e^t \sin s, \quad y = 3e^t \cos s, \quad z = 4e^t.$$

Find $\frac{\partial w}{\partial s}$ at $s = t = 0$.

Solution: By the chain rule in classical notation, we have

$$\begin{aligned}\frac{\partial w}{\partial s} &= \frac{\partial w}{\partial x} \cdot \frac{\partial x}{\partial s} + \frac{\partial w}{\partial y} \cdot \frac{\partial y}{\partial s} + \frac{\partial w}{\partial z} \cdot \frac{\partial z}{\partial s} \\ &= \frac{x}{\sqrt{x^2 + y^2 + z^2}} \cdot 3e^t \cos s - \frac{y}{\sqrt{x^2 + y^2 + z^2}} \cdot 3e^t \sin s.\end{aligned}$$

As $s = t = 0$, $(x, y, z) = (0, 3, 4)$ and hence

$$\frac{\partial w}{\partial s} \Big|_{(s,t)=(0,0)} = 0.$$

Example 101. Roy is walking with position at time t given by

$$x(t) = t^3 + 1, \quad y(t) = 2t^2.$$

His altitude is $H(x, y) = x^2 - y^2 + 100$.

- Is Roy going uphill or downhill at $t = 1$?
- Which direction should he go instead at time $t = 1$ to move downhill the most quickly?

Solution:

- We must find $\frac{\partial H}{\partial t} \Big|_{t=1}$. By the Chain rule

$$\begin{aligned}\frac{\partial H}{\partial t} &= \frac{\partial H}{\partial x} \cdot \frac{\partial x}{\partial t} + \frac{\partial H}{\partial y} \cdot \frac{\partial y}{\partial t} \\ &= (2x)(3t^2) + (-2y)(4t) \\ &= 2(t^3 + 1)(3t^2) - 4t^2(4t) \\ &= 6t^5 - 16t^3 + 6t^2.\end{aligned}$$

Therefore

$$\frac{\partial H}{\partial t} \Big|_{t=1} = 6 - 16 + 6 = -4 < 0,$$

and Roy is going downhill at $t = 1$.

- At $t = 1$, $(x, y) = (2, 2)$, and $\nabla H(x, y) = (2x, -2y)$. So at time $t = 1$, the gradient of H is given by $\nabla H(2, 2) = (4, -4)$. Therefore H decreases most rapidly in the direction of $-\nabla H(2, 2)$, which is given by the vector

$$u = \left(-\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right).$$

That is, Roy should travel in the direction u , which if $(1, 0)$ points east and $(0, 1)$ points north, is northwest.

Note that in the previous example

$$\frac{\partial H}{\partial t} = \nabla H \cdot \left(\frac{\partial x}{\partial t}, \frac{\partial y}{\partial t}\right),$$

which in words can be stated as the rate of change of the altitude is given by the gradient of the altitude dotted with the velocity you are travelling.

4.3.1 Operator Norm (non-examinable)

Before proving the chain rule, we make a brief detour regarding the operator norm. Let $A : \mathbb{R}^n \rightarrow \mathbb{R}^m$ be a linear map and for a non-zero vector $\vec{x} \in \mathbb{R}^n$, consider the quantity

$$\frac{\|A\vec{x}\|}{\|\vec{x}\|} \in \mathbb{R},$$

which is the length of the vector $A\vec{x} \in \mathbb{R}^m$ divided by the length of the vector $\vec{x} \in \mathbb{R}^n$. Note that since A is linear,

$$A\left(\frac{\vec{x}}{\|\vec{x}\|}\right) = \frac{A(\vec{x})}{\|\vec{x}\|},$$

and so

$$\left\|A\left(\frac{\vec{x}}{\|\vec{x}\|}\right)\right\| = \frac{\|A\vec{x}\|}{\|\vec{x}\|}.$$

Therefore, we may assume that $\vec{x}$ is a unit vector and consider the quantity $\|A(\vec{x})\| \in \mathbb{R}$ instead.

For any unit vector $\vec{x} \in \mathbb{R}^n$, we note that the i^{th} -component of $A\vec{x}$ (for some $1 \leq i \leq m$) satisfies the inequality

$$|(A\vec{x})_i| = |\alpha_i \cdot \vec{x}| \leq \|\alpha_i\| \cdot \underbrace{\|\vec{x}\|}_{=1} = \|\alpha_i\|,$$

where α_i denotes the i^{th} -row of A as before. In particular, we have

$$\|A\vec{x}\|^2 = \sum_{i=1}^m (A\vec{x})_i^2 \leq \sum_{i=1}^m \|\alpha_i\|^2 = \sum_{i=1}^m \sum_{j=1}^n a_{ij}^2 = \|A\| < \infty,$$

where the right hand side is the length of A when considered as a vector in the vector space $\mathbb{R}^{nm}$. Hence we have a uniform upper bound on our quantity for any unit vector $\vec{x} \in \mathbb{R}^n$. This leads to the following definition.

Definition 4.21. Let $A : \mathbb{R}^n \rightarrow \mathbb{R}^m$ be a linear map. The operator norm of A is defined to be

$$\|A\|_{op} := \sup_{\vec{x} \in \mathbb{R}^n \setminus \{0\}} \frac{\|A\vec{x}\|}{\|\vec{x}\|} = \sup_{\|\vec{x}\|=1} \|A\vec{x}\|.$$

Remark. Directly from the definition we find that the inequality

$$\|A\vec{x}\| \leq \|A\|_{op} \|\vec{x}\|,$$

holds for every $\vec{x} \in \mathbb{R}^n$.

Example 102. Consider the matrix $A = \begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix}$. Note that for any unit vector $\vec{v}$, we have

$$A\vec{v} = \begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 2x \\ y \end{pmatrix},$$

and so

$$\|A\vec{v}\| = \sqrt{4x^2 + y^2} \leq \sqrt{4x^2 + 4y^2} = 2,$$

or $\|A\|_{op} \leq 2$. However, since $\|A(\vec{e}_1)\| = 2$, we can conclude that $\|A\|_{op} = 2$.

Exercise: Suppose A is a $n \times n$ matrix with real eigenvalues $\lambda_1, \dots, \lambda_n \in \mathbb{R}$. What is the operator norm $\|A\|_{op}$ in terms of the eigenvalues?

4.3.2 Proof of the Chain Rule

Proof of Chain Rule. Since $\vec{f}$ is differentiable at $\vec{a}$:

$$\vec{f}(\vec{x}) - \vec{f}(\vec{a}) = D\vec{f}(\vec{a})(\vec{x} - \vec{a}) + \varepsilon_{\vec{f}}(\vec{x}), \quad \forall \vec{x} \in \Omega_1, \quad (4.15)$$

with $\lim_{\vec{x} \rightarrow \vec{a}} \frac{\|\varepsilon_{\vec{f}}(\vec{x})\|}{\|\vec{x} - \vec{a}\|} = 0$. Since $\vec{g}$ is differentiable at $\vec{b}$:

$$\vec{g}(\vec{y}) - \vec{g}(\vec{b}) = D\vec{g}(\vec{b})(\vec{y} - \vec{b}) + \varepsilon_{\vec{g}}(\vec{y}), \quad \forall \vec{y} \in \Omega_2, \quad (4.16)$$

with $\lim_{\vec{y} \rightarrow \vec{b}} \frac{\|\varepsilon_{\vec{g}}(\vec{y})\|}{\|\vec{y} - \vec{b}\|} = 0$. Setting $\vec{y} = \vec{f}(\vec{x})$, $\vec{b} = \vec{f}(\vec{a})$, and substituting (4.15) into (4.16)

$$\begin{aligned} \vec{g} \circ \vec{f}(\vec{x}) - \vec{g} \circ \vec{f}(\vec{a}) &= D\vec{g}(\vec{f}(\vec{a})) \left(D\vec{f}(\vec{a})(\vec{x} - \vec{a}) + \varepsilon_{\vec{f}}(\vec{x}) \right) + \varepsilon_{\vec{g}}(\vec{f}(\vec{x})) \\ &= \underbrace{D\vec{g}(\vec{f}(\vec{a})) \cdot D\vec{f}(\vec{a})(\vec{x} - \vec{a})}_{D(\vec{g} \circ \vec{f})(\vec{a})} + \underbrace{D\vec{g}(\vec{f}(\vec{a}))\varepsilon_{\vec{f}}(\vec{x}) + \varepsilon_{\vec{g}}(\vec{f}(\vec{x}))}_{\varepsilon_{\vec{g} \circ \vec{f}}(\vec{x})}. \end{aligned}$$

Therefore, to finish the proof, it suffices to show that

$$\lim_{\vec{x} \rightarrow \vec{a}} \frac{\|\varepsilon_{\vec{g} \circ \vec{f}}(\vec{x})\|}{\|\vec{x} - \vec{a}\|} = 0.$$

In order to do this, by the triangle inequality

$$\frac{\|\varepsilon_{\vec{g} \circ \vec{f}}(\vec{x})\|}{\|\vec{x} - \vec{a}\|} \leq \frac{\|D\vec{g}(\vec{f}(\vec{a}))\varepsilon_{\vec{f}}(\vec{x})\|}{\|\vec{x} - \vec{a}\|} + \frac{\|\varepsilon_{\vec{g}}(\vec{f}(\vec{x}))\|}{\|\vec{x} - \vec{a}\|},$$

so if the limit as $\vec{x} \rightarrow \vec{a}$ of both terms on the right hand side is zero, then we are done by the Squeeze theorem. We first note that

$$\|D\vec{g}(\vec{f}(\vec{a})) \cdot \varepsilon_{\vec{f}}(\vec{x})\| \leq \|D\vec{g}(\vec{f}(\vec{a}))\|_{op} \cdot \|\varepsilon_{\vec{f}}(\vec{x})\|,$$

and since $\vec{f}$ is differentiable at $\vec{a}$,

$$\lim_{\vec{x} \rightarrow \vec{a}} \frac{\|D\vec{g}(\vec{f}(\vec{a}))\|_{op} \|\varepsilon_{\vec{f}}(\vec{x})\|}{\|\vec{x} - \vec{a}\|} = \|D\vec{g}(\vec{f}(\vec{a}))\|_{op} \cdot \lim_{\vec{x} \rightarrow \vec{a}} \frac{\|\varepsilon_{\vec{f}}(\vec{x})\|}{\|\vec{x} - \vec{a}\|} = 0.$$

Therefore by the Squeeze theorem

$$\lim_{\vec{x} \rightarrow \vec{a}} \frac{\|D\vec{g}(\vec{f}(\vec{a})) \cdot \varepsilon_{\vec{f}}(\vec{x})\|}{\|\vec{x} - \vec{a}\|} = 0.$$

Contents

Next, we note that if $\vec{f}(\vec{x}) = \vec{f}(\vec{a}) = \vec{b}$, then since $\varepsilon_{\vec{g}}(\vec{b}) = 0$, the quantity we want to control $\frac{\|\varepsilon_{\vec{g}}(\vec{f}(\vec{x}))\|}{\|\vec{x} - \vec{a}\|}$ is identically zero, and there is nothing to show. Thus, without loss of generality, we may assume that $\vec{f}(\vec{x}) \neq \vec{f}(\vec{a})$ and hence rewrite the quantity as

$$\frac{\|\varepsilon_{\vec{g}}(\vec{f}(\vec{x}))\|}{\|\vec{x} - \vec{a}\|} = \frac{\|\varepsilon_{\vec{g}}(\vec{f}(\vec{x}))\|}{\|\vec{f}(\vec{x}) - \vec{f}(\vec{a})\|} \cdot \frac{\|\vec{f}(\vec{x}) - \vec{f}(\vec{a})\|}{\|\vec{x} - \vec{a}\|}.$$

Since for $\vec{x}$ near $\vec{a}$, we can bound the quantity

$$\begin{aligned} \frac{\|\vec{f}(\vec{x}) - \vec{f}(\vec{a})\|}{\|\vec{x} - \vec{a}\|} &= \frac{\|D\vec{f}(\vec{a})(\vec{x} - \vec{a}) + \varepsilon_{\vec{f}}(\vec{x})\|}{\|\vec{x} - \vec{a}\|} \\ &\leq \frac{\|D\vec{f}(\vec{a})(\vec{x} - \vec{a})\|}{\|\vec{x} - \vec{a}\|} + \frac{\|\varepsilon_{\vec{f}}(\vec{x})\|}{\|\vec{x} - \vec{a}\|} \\ &\leq \|D\vec{f}(\vec{a})\|_{op} + 1 < \infty, \end{aligned}$$

and as

$$\lim_{\vec{x} \rightarrow \vec{a}} \frac{\|\varepsilon_{\vec{g}}(\vec{f}(\vec{x}))\|}{\|\vec{f}(\vec{x}) - \vec{f}(\vec{a})\|} = \lim_{\vec{y} \rightarrow \vec{b}} \frac{\|\varepsilon_{\vec{g}}(\vec{y})\|}{\|\vec{y} - \vec{b}\|} = 0,$$

it follows that

$$\lim_{\vec{x} \rightarrow \vec{a}} \frac{\|\varepsilon_{\vec{g}}(\vec{f}(\vec{x}))\|}{\|\vec{x} - \vec{a}\|} = 0. \quad \square$$

4.3.3 Summary

(i) $f : \Omega \subseteq \mathbb{R} \rightarrow \mathbb{R}$ (one variable, real-valued).

$$Df(x) = \frac{df}{dx} \quad (1 \times 1 \text{ matrix})$$

(ii) $f : \Omega \subseteq \mathbb{R}^n \rightarrow \mathbb{R}$ (multivariable, scalar-valued).

$$Df(\vec{x}) = \nabla f(\vec{x}) = \left(\frac{\partial f}{\partial x_1}(\vec{x}), \dots, \frac{\partial f}{\partial x_n}(\vec{x}) \right) \quad (1 \times n \text{ matrix})$$

(iii) $f : \Omega \subseteq \mathbb{R}^n \rightarrow \mathbb{R}^m$ (multivariable, vector-valued).

$$D\vec{f}(\vec{x}) = \begin{pmatrix} -\nabla f_1(\vec{x}) - \\ \vdots \\ -\nabla f_m(\vec{x}) - \end{pmatrix} = \left(\frac{\partial f_i}{\partial x_j}(\vec{x}) \right)_{i,j} \quad (m \times n \text{ matrix})$$

Chain Rule: Consider functions $f : \Omega_1 \subset \mathbb{R}^k \rightarrow \mathbb{R}^n$, $g : \Omega_2 \subset \mathbb{R}^n \rightarrow \mathbb{R}^m$. The chain rule states that, at a point $\vec{a} \in \Omega_1$,

$$D(\vec{g} \circ \vec{f})(\vec{a}) = D\vec{g}(\vec{f}(\vec{a})) \cdot D\vec{f}(\vec{a})$$

Contents

Consider the above in coordinates:

$$(x_1, \dots, x_n) \xrightarrow{f} (y_1, \dots, y_m) \xrightarrow{g} (g_1, \dots, g_k),$$

$g_i = g_i(y_1, \dots, y_m)$ is a function of the variables $y_1, \dots, y_m$,

$y_j = f_j(x_1, \dots, x_n)$ is a change of variables from the x_i to the y_j .

Then, with an abuse of notation, the components of the composition $g \circ f$ can be considered as $g_i = g_i(x_1, \dots, x_n)$; functions of the variables $x_1, \dots, x_n$. With this set-up, the Chain Rule becomes

$$\begin{pmatrix} \frac{\partial g_1}{\partial x_1} & \dots & \frac{\partial g_1}{\partial x_n} \\ \vdots & \ddots & \vdots \\ \frac{\partial g_k}{\partial x_1} & \dots & \frac{\partial g_k}{\partial x_n} \end{pmatrix} = \begin{pmatrix} \frac{\partial g_1}{\partial y_1} & \dots & \frac{\partial g_1}{\partial y_m} \\ \vdots & \ddots & \vdots \\ \frac{\partial g_k}{\partial y_1} & \dots & \frac{\partial g_k}{\partial y_m} \end{pmatrix} \cdot \begin{pmatrix} \frac{\partial y_1}{\partial x_1} & \dots & \frac{\partial y_1}{\partial x_n} \\ \vdots & \ddots & \vdots \\ \frac{\partial y_m}{\partial x_1} & \dots & \frac{\partial y_m}{\partial x_n} \end{pmatrix}$$

Looking at the i^{th} row and j^{th} column of the matrix on the left hand side, we find that

$$\frac{\partial g_i}{\partial x_j} = \left(\frac{\partial g_i}{\partial y_1}, \dots, \frac{\partial g_i}{\partial y_m} \right) \cdot \begin{pmatrix} \frac{\partial y_1}{\partial x_j} \\ \vdots \\ \frac{\partial y_m}{\partial x_j} \end{pmatrix} = \sum_{a=1}^m \frac{\partial g_i}{\partial y_a} \cdot \frac{\partial y_a}{\partial x_j}.$$